

Python Basics Practice Assignment

Objective

The goal of this assignment is to help students become familiar with the **basics of Python programming**, including comments, variables, operators, input/output, type casting, and conditional statements.

A. Comments and Variables

1. Write a Python program that stores your name in a variable and prints it.

2. Identify whether the following variable names are valid or invalid:

- `1student` – Invalid (cannot start with a number)
- `_student` – Valid
- `student_name` – Valid
- `full name` – Invalid (spaces not allowed)
- `and` – Invalid (reserved keyword)

3. Correct the following code:

```
1 my age = 20
2 print(my age)
```

4. Create meaningful variable names for:

- Your age
- Number of students in a class
- Price of a product

5. Write a program that assigns values to three variables and prints them using a single `print()` statement.

6. **Theory: Why should variable names be meaningful?**

Answer: Meaningful variable names make the program easier to read, understand, and maintain. They help programmers quickly understand what data is being stored.

B. Arithmetic Operators

7. Write a program to add, subtract, multiply, and divide two numbers.

8. What will be the output of the following code?

```
1 x = 10
2 y = 3
3
4 print(x // y)
5 print(x % y)
```

9. Write a program to calculate the square and cube of a number.

10. What will be the output?

```
1 print(5 + 10 / 2 ** 2)
```

11. Write a program to calculate the total price if:

- Price of one item = 120
- Quantity = 7

12. Write a program to calculate the average of three numbers.

13. Write a program that calculates power using the ****** operator.

C. Input and Type Casting

14. Write a program that takes an integer input from the user and prints its double.

15. Write a program that takes one integer and one float input and prints their sum.

16. **Theory: What happens if we do not use `int()` or `float()` with `input()`?**

Answer: The `input()` function always returns a string. Without type casting, mathematical operations may produce incorrect results or errors.

17. Write a program that asks the user for their birth year and calculates their age.

18. Write a program that takes length and breadth as input and calculates the area of a rectangle.

D. Real-Life Conversion Problems

19. Write a program to convert:
- Kilometer to meter
 - Meter to kilometer
20. Write a program to convert minutes into seconds.
21. Write a program to convert Celsius to Fahrenheit using:

$$F = (C \times \frac{9}{5}) + 32$$

22. Write a program to convert Nepali Rupees into Paisa.

E. Relational Operators and Conditions

23. What will be the output?

```
1 print(10 > 5)
2 print(10 == 5)
3 print(10 != 5)
```

24. Write a program that checks whether a number is positive or negative.
25. Write a program that checks whether a number is even or odd.
26. Write a program that takes age as input and checks if the user is eligible to vote.
27. Write a program that checks:
- Marks $\geq 40 \rightarrow$ Pass
 - Else $\rightarrow$ Fail
28. Write a program that checks the greater number between two numbers.
29. Write a program that checks citizenship eligibility:
- Male $\geq 18 \rightarrow$ Eligible
 - Female $\geq 16 \rightarrow$ Eligible

30. **Theory:** What is indentation in Python and why is it important?

Answer: Indentation refers to the spaces at the beginning of a line of code. Python uses indentation to define code blocks. Incorrect indentation causes errors or incorrect program execution.

End of Assignment